

Mathematics for Machine Learning

MAI391

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1. Analytic Geometry

1.1 Norm

Definition 1.1.1

A norm on V is a non-negative function

$$\|\cdot\| : V \rightarrow \mathbb{R} \quad (1.1)$$

such that $\forall k \in \mathbb{R}$ and $x, y \in V$:

- *Absolute homogeneous*: $\|kx\| = |k|\|x\|$
- **Triangle inequality**: $\|x + y\| \leq \|x\| + \|y\|$
- **Positive definite**: $\|x\| \geq 0$ and $\|x\| = 0$ if and only if $x = \vec{0}$



Definition 1.1.2

Let p be as positive number. The l_p -norm $\|\cdot\|_p$ in $\mathbb{R}^n$:

$$\|x\|_p = \sqrt[p]{\sum_{i=1}^n |x_i|^p} \quad \text{for any } x = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} \in \mathbb{R}^n \quad (1.2)$$



Example 1.1.3

- $p = 1 \rightarrow$ Manhattan norm l_1 :

$$\|x\|_1 = \sum_{i=1}^n |x_i| \quad (1.3)$$

- $p = 2 \rightarrow$ Euclidean norm l_2 :

$$\|x\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2} \quad (1.4)$$

- $p = \infty \rightarrow$ maximum norm l_∞ :

$$\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\} \quad (1.5)$$



Definition 1.1.4

Euclidean norm on $\mathbb{R}^{m \times n}$:

$$\|A\| = \left(\sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \right)^{\frac{1}{2}} \quad (1.6)$$

**Definition 1.1.5**

Spectral norm on $\mathbb{R}^{m \times n}$:

$$\|A\|_2 = \max_{x \in \mathbb{R}^n, \|x\|_2=1} \|Ax\|_2 \quad (1.7)$$



1.2 Inner Product

Definition 1.2.1

A linear mapping L on V is a function $L : V \rightarrow \mathbb{R}$ that satisfies:

- $L(kx) = kL(x)$
- $L(x + y) = L(x) + L(y)$

$\forall x \in V$ and $k \in \mathbb{R}$

**Definition 1.2.2**

A bilinear mapping or bilinear form B on V is a function $B : V \times V \rightarrow \mathbb{R}$ that is **linear in each argument separately**:

- $B(kx, z) = kB(x, z)$
- $B(x + y, z) = B(x, z) + B(y, z)$
- $B(x, kz) = kB(x, z)$
- $B(x, y + z) = B(x, y) + B(x, z)$

$\forall x, y, z \in V$ and $k \in \mathbb{R}$



Problem 1.2.3

Show that the mapping $B : R^2 \times R^2 \rightarrow R$

$$B(x, y) = x_1 y_1 - 2x_1 y_2 + 3x_2 y_2 \quad (1.8)$$

is a bilinear mapping on R^2 .

◇

Solution.

$$\begin{aligned} B(kx, z) &= kx_1 z_1 - 2kx_1 z_2 + 3kx_2 z_2 = k(x_1 z_1 - 2x_1 z_2 + 3x_2 z_2) = kB(x, z) \\ B(x + y, z) &= (x_1 + y_1)z_1 - 2(x_1 + y_1)z_2 + 3(x_2 + y_2)z_2 \\ &= (x_1 z_1 - 2x_1 z_2 + 3x_2 z_2) + (y_1 z_1 - 2y_1 z_2 + 3y_2 z_2) \\ &= B(x, z) + B(y, z) \\ B(x, kz) &= kx_1 z_1 - 2kx_1 z_2 + 3kx_2 z_2 = k(x_1 z_1 - 2x_1 z_2 + 3x_2 z_2) = kB(x, z) \\ B(x, y + z) &= x_1(y_1 + z_1) - 2x_1(y_2 + z_2) + 3x_2(y_2 + z_2) \\ &= (x_1 y_1 - 2x_1 y_2 + 3x_2 y_2) + (x_1 z_1 - 2x_1 z_2 + 3x_2 z_2) \\ &= B(x, y) + B(x, z) \end{aligned}$$

$\Rightarrow B$ is linear mapping

□

Definition 1.2.4

A bilinear mapping $B : V \times V \rightarrow R$ is called

- **symmetric** if $B(x, y) = B(y, x)$
- **positive definite** if $B(x, x) > 0$ for any $V \ni x \neq \vec{0}$

♣

Definition 1.2.5

A positive definite, symmetric bilinear mapping $B : V \times V \rightarrow R$ is called an **inner product**. We typically write $\langle x, y \rangle$ instead of $B(x, y)$ for inner product

♣

Definition 1.2.6

The pair $(V, \langle \cdot, \cdot \rangle)$ is called **inner product space**

♣

Definition 1.2.7

The **dot product** or **scalar product** on $\mathbb{R}^n$ is given by

$$x^T y = \sum_{i=1}^n x_i y_i \quad (1.10)$$

**Property 1.2.8**

Dot product is inner product, and $(\mathbb{R}^n, \langle \cdot, \cdot \rangle)$ is **Euclidean vector space**

**Problem 1.2.9**

Consider $V = \mathbb{R}^2$. Define

$$\langle x, y \rangle = 2x_1 y_1 + x_1 y_2 + x_2 y_1 + 2x_2 y_2 \quad (1.11)$$

Show that $\langle \cdot, \cdot \rangle$ is an inner product in $\mathbb{R}^2$



Solution.

- $\langle \cdot, \cdot \rangle$ is a bilinear mapping (*The solution is similar to Problem 1.2.3*)
- $\langle \cdot, \cdot \rangle$ is symmetric

$$\begin{aligned} \langle \cdot, \cdot \rangle &= 2x_1 y_1 + x_1 y_2 + x_2 y_1 + 2x_2 y_2 \\ &= 2y_1 x_1 + y_1 x_2 + y_2 x_1 + 2x_2 y_2 = B(y, x) \end{aligned} \quad (1.12)$$

- $\langle \cdot, \cdot \rangle$ is positive definite

$$\begin{aligned} \langle \cdot, \cdot \rangle &= 2x_1^2 + 2x_2^2 + 2x_1 x_2 \\ &= (x_1 + x_2)^2 + x_1^2 + x_2^2 \geq 0 \end{aligned} \quad (1.13)$$

The equality occur only when $x_1 = 0, x_2 = 0, x_1 + x_2 = 0$ which is $x = \vec{0}$
 $\Rightarrow \langle \cdot, \cdot \rangle > 0$ for any $\mathbb{R}^2 \ni x \neq \vec{0}$, or $\langle \cdot, \cdot \rangle$ is positive definite

So $\langle \cdot, \cdot \rangle$ satisfies all properties to be an inner product



Definition 1.2.10

A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is

- **positive definite** if

$$x^T A x > 0, \text{ for all non-zero vectors } x \in \mathbb{R}^n \quad (1.14)$$

- **positive semidefinite** if

$$x^T A x \geq 0, \text{ for all non-zero vectors } x \in \mathbb{R}^n \quad (1.15)$$

**Problem 1.2.11**

Consider the matrices

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \quad (1.16)$$

Show that A is positive definite but B is not



Solution.

$$\begin{aligned} x^T A x &= [x_1 \ x_2] \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= 2x_1^2 + 2x_1x_2 + 3x_2^2 \\ &> 0 \text{ for all non-zero vectors } x \in \mathbb{R}^n \end{aligned} \quad (1.17)$$

(the proof for why > 0 is similar to [Problem 1.2.9](#))

Also A is symmetric so A is positive definite

$$\begin{aligned} x^T B x &= [x_1 \ x_2] \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= x_1^2 + 4x_1x_2 + 3x_2^2 \end{aligned} \quad (1.18)$$

Take $x = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \Rightarrow x^T B x = -1$

So B is not positive definite and not positive semidefinite



Note 1.2.12

When multiply matrices of form $x^T Ax$ (like in [Problem 1.2.11](#)), the coefficient of

$$x_i x_j = \begin{cases} a_{ij}, & \text{if } i = j \\ a_{ij} + a_{ji}, & \text{if } i \neq j \end{cases} \quad (1.19)$$



1.3 Lengths and Distances

Definition 1.3.1

The **norm** is $\|x\| := \sqrt{\langle x, x \rangle}$

**Definition 1.3.2**

The **distance between x and y** is $d(x, y) := \|x - y\| = \sqrt{\langle x - y, x - y \rangle}$

**Theorem 1.3.3**

Cauchy-Schwarz inequality

$$|\langle x, y \rangle| \leq \|x\| \|y\| \quad (1.20)$$

**Problem 1.3.4**

Let the inner product $\langle \cdot, \cdot \rangle$ be defined on $\mathbb{R}^n$ by

$$\langle x, y \rangle = x_1 y_1 - x_1 y_2 - x_2 y_1 + 3x_2 y_2 \quad (1.21)$$

Find the distance between $u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $v = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$



Solution.

$$u - v = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad (1.22)$$

$$\begin{aligned}
d(u, v) &= \|u - v\| = \sqrt{\langle u - v, u - v \rangle} \\
&= \sqrt{2 \cdot 2 - 2 \cdot (-1) - (-1) \cdot 2 + 3 \cdot (-1) \cdot (-1)} \\
&= \sqrt{11}
\end{aligned} \tag{1.23}$$

□

1.4 Angles and Orthogonality

Definition 1.4.1

The **angle** between vector x and y is the number $\theta \in [0, \pi]$ defined by

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} \tag{1.24}$$

♣

Problem 1.4.2

Define the inner product $\langle \cdot, \cdot \rangle$

$$\langle x, y \rangle = x \cdot y = x_1 y_1 + x_2 y_2 \text{ (dot product)} \tag{1.25}$$

Find the angle between $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $y = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$

◇

Solution.

$$\langle x, y \rangle = 1 \cdot \left(-\frac{1}{2}\right) + 0 \cdot \frac{\sqrt{3}}{2} = -\frac{1}{2} \tag{1.26}$$

$$\begin{aligned}
\|x\| &= \sqrt{1 \cdot 1 + 0 \cdot 0} = 1 \\
\|y\| &= \sqrt{-\frac{1}{2} \cdot -\frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2}} = 1
\end{aligned} \tag{1.27}$$

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} = \frac{-\frac{1}{2}}{1 \cdot 1} = -\frac{1}{2} \tag{1.28}$$

$$\Rightarrow \theta = \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3} \tag{1.29}$$

□

Definition 1.4.3

Two vectors x and y are **orthogonal** (or $x \perp y$) $\Leftrightarrow \langle x, y \rangle = 0$

**Definition 1.4.4**

If $x \perp y$ and $\|x\| = \|y\| = 1$ (x and y are unit vectors) then x and y are **orthonormal**

**Definition 1.4.5**

A square matrix $A \in \mathbb{R}^{n \times n}$ is an **orthogonal matrix** if and only if

$$AA^T = A^T A = I_n \quad (1.30)$$

**Property 1.4.6**

If $A \in \mathbb{R}^{n \times n}$ is orthogonal then

- $A^{-1} = A^T$
- A preserve the length of any vector $x \in \mathbb{R}^n$

$$\|Ax\| = \|x\| \quad (1.31)$$

- A preserve the angle between two vectors x and y in $\mathbb{R}^n$

$$\text{angle}(Ax, Ay) = \text{angle}(x, y) \quad (1.32)$$

**Note 1.4.7**

For square matrix $A \in \mathbb{R}^{n \times n}$, A^{-1} is defined as matrix with

$$AA^{-1} = A^{-1}A = I_n \quad (1.33)$$

So if A is orthogonal, $A^T = A^{-1}$



1.5 Orthonormal Basis

Definition 1.5.1

Given a vector space V , a set of vectors $\{v_1, \dots, v_n\}$ is **linear independent** if the **only solution** to equation

$$a_1 v_1 + \dots + a_n v_n = 0 \quad (1.34)$$

is $a_1 = \dots = a_n = 0$



Definition 1.5.2

Given a vector space V , a set of vectors $B = \{b_1, \dots, b_n\}$ is basis if:

- **Every subset** of B is **linear independent**
- For every $x \in V$ there exist $\{a_1, \dots, a_n\}$ such that

$$a_1 b_1 + \dots + a_n b_n = x \quad (1.35)$$



Example 1.5.3

Most common basis for $\mathbb{R}^2$ is $B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$



Definition 1.5.4

Consider an dimensional inner product vector space $(V, \langle \cdot, \cdot \rangle)$ and a basis $B = \{b_1, \dots, b_n\}$ of V . B is **orthonormal basis** if

$$\langle b_i, b_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (1.36)$$



Theorem 1.5.5

Let $B = \{b_1, \dots, b_n\}$ be an orthonormal basis of an inner product space V . If x is any vector in V , then

$$x = \langle x, b_1 \rangle b_1 + \dots + \langle x, b_n \rangle b_n \quad (1.37)$$



Problem 1.5.6

Show that $B = \left\{ \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ is an orthonormal basis of $\mathbb{R}^n$ with inner product

$$\langle x, y \rangle = x^T A y, \text{ where } A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \quad (1.38)$$

◇

Solution.

$$\begin{aligned} \langle b_1, b_1 \rangle &= \left\langle \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\rangle \\ &= [1 \ -1] A \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= 1 \end{aligned} \quad (1.39)$$

$$\begin{aligned} \langle b_1, b_2 \rangle &= \left\langle \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\rangle \\ &= [1 \ -1] A \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= 0 \end{aligned} \quad (1.40)$$

$$\begin{aligned} \langle b_1, b_2 \rangle &= \left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\rangle \\ &= [1 \ 0] A \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= 0 \end{aligned} \quad (1.41)$$

$$\begin{aligned} \langle b_2, b_2 \rangle &= \left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\rangle \\ &= [1 \ 0] A \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ &= 1 \end{aligned} \quad (1.42)$$

So B is an orthonormal basis

Theorem 1.5.7 (Gram-Schmidt process)

Let V be an inner product space and let $\{v_1, \dots, v_n\}$ be **any basis** of V . Define vectors $b_1, b_2, \dots, b_n$ in V as follows:

$$\begin{aligned} b_1 &= v_1 \\ b_2 &= v_2 - \frac{\langle v_2, b_1 \rangle}{\|b_1\|^2} b_1 \\ &\dots \\ b_k &= v_k - \frac{\langle v_k, b_1 \rangle}{\|b_1\|^2} b_1 - \frac{\langle v_k, b_2 \rangle}{\|b_2\|^2} b_2 - \dots - \frac{\langle v_k, b_{k-1} \rangle}{\|b_{k-1}\|^2} b_{k-1} \end{aligned} \quad (1.43)$$

for each $k = 2, \dots, n$. Then

- $\{b_1, \dots, b_n\}$ is an orthogonal basis of V
- $\text{span}\{b_1, \dots, b_k\} = \text{span}\{v_1, \dots, v_k\}$ for each $k = 1, \dots, n$
- $\left\{ \frac{b_1}{\|b_1\|}, \dots, \frac{b_n}{\|b_n\|} \right\}$ is an orthonormal basis of V

**Problem 1.5.8**

Given basis

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\} \quad (1.44)$$

and inner product as **dot product**.

Apply Gram-Schmidt process to find orthonormal basis



Solution.

$$\begin{aligned} b_1 &= v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\ \Rightarrow e_1 &= \frac{b_1}{\|b_1\|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} \end{aligned} \quad (1.45)$$

$$\begin{aligned}
b_2 &= v_2 - \frac{\langle v_2, b_1 \rangle}{\|b_1\|^2} b_1 \\
&= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{\begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}}{\left\| \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\|^2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \\
&= \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix} \\
\Rightarrow e_2 &= \frac{b_2}{\|b_2\|} = \left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}} \right)
\end{aligned} \tag{1.46}$$

$$\begin{aligned}
b_3 &= v_3 - \frac{\langle v_3, b_1 \rangle}{\|b_1\|^2} b_1 - \frac{\langle v_3, b_2 \rangle}{\|b_2\|^2} b_2 = \begin{bmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix} \\
\Rightarrow e_3 &= \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}
\end{aligned} \tag{1.47}$$

So we have **orthogonal basis**

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}, \begin{bmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix} \right\} \tag{1.48}$$

and **orthonormal basis**

$$\left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \right\} \tag{1.49}$$

□

2. Matrix Decomposition

2.1 Determinant

Example 2.1.1

$$A = [a] \in \mathbb{R}^{1 \times 1} \Rightarrow \det(A) = a \quad (2.1)$$

Example 2.1.2

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \in \mathbb{R}^{2 \times 2} \Rightarrow \det(A) = ad - bc \quad (2.2)$$

Property 2.1.3

For $A \in \mathbb{R}^{n \times n}$

$$\det(AB) = \det(A) \det(B) \quad (2.3)$$

$$\det(A^T) = \det(A) \quad (2.4)$$

$$\det(A^{-1}) = \frac{1}{\det(A)} \quad (2.5)$$

$$\det(kA) = k^n \det(A) \quad (2.6)$$

Property 2.1.4

- Adding a multiple of a column/row to another one (i.e., apply $R_2 = R_2 - 2 * R_1$) does not change $\det(A)$
- Swapping two rows/columns changes the sign of $\det(A)$ (i.e., $\det(A) = -\det(A)$ when swapping row 1 and 2)

Theorem 2.1.5

Let $A \in \mathbb{R}^{n \times n}$. The followings are equivalent (mean that if one is correct then all others are correct, for example if 1 is correct then 2, 3, 4, 5 is correct):

- $\det(A) \neq 0$
- A has full rank. $\text{rk}(A) = n$
- A is invertible (A^{-1} exist)
- The homogeneous system $Ax = 0$ for $x \in \mathbb{R}^n$ has only trivial solution (the solution is $x = 0 = [0 \ \dots \ 0]^T$)
- The linear system $Ax = b$ for $x, b \in \mathbb{R}^n$ has unique solution

**Note 2.1.6**

Rank of a matrix is equal to number of pivot in its row echelon form

**2.2 Trace****Definition 2.2.1**

The **trace** of a matrix $A \in \mathbb{R}^{n \times n}$ is defined as:

$$\text{tr}(A) = \sum_{i=1}^n a_{ii} \quad (2.7)$$

, the sum of the diagonal elements of A

**Example 2.2.2**

$$\text{tr} \left(\begin{bmatrix} 2 & 3 & 1 \\ 0 & -3 & 4 \\ 1 & -2 & 4 \end{bmatrix} \right) = 2 - 3 + 4 = 3 \quad (2.8)$$



Property 2.2.3

$$\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B) \quad (2.9)$$

$$\text{tr}(kA) = k \text{tr}(A) \quad (2.10)$$

$$\text{tr}(I_n) = n \quad (2.11)$$

$$\text{tr}(AB) = \text{tr}(BA) \quad (2.12)$$



2.3 Dimension

For later

2.4 Eigenvalues and Eigenvectors

Definition 2.4.1

The **characteristic polynomial** of $A \in \mathbb{R}^{n \times n}$

$$p_A := \det(A - \lambda I_n) = c_0 + c_1 \lambda + \dots c_n \lambda^n \quad (2.13)$$

Where:

- $c_0 = \det(A)$
- $c_{n-1} = (-1)^{n-1} \text{tr}(A)$
- $c_n = (-1)^n$

**Definition 2.4.2**

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix. Then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A and non-zero vector $x \in \mathbb{R}^n$ is the **corresponding eigenvector** of A if

$$Av = \lambda v \quad (2.14)$$



Theorem 2.4.3

The following statements are equivalent:

- λ is an eigenvalue of A
- There exists a non-zero vector $v \in \mathbb{R}^n$ with $Av = \lambda v$
- The homogeneous system $(A - \lambda I_n)v = 0$ has non trivial solution
- $\text{rk}(A - \lambda I_n) < n$ (*not full rank*)
- $\det(A - \lambda I_n) = 0$
- λ is a root of the **characteristic polynomial** $p_A(\lambda)$ of A (i.e., $p_A(\lambda) = 0$)

**Definition 2.4.4**

Let a square matrix A have an eigenvalue λ_i , then:

- The **algebraic multiplicity** of λ_i is the number of times the root appears in the characteristic polynomial
- The **eigenspace** of A w.r.t λ_i , denoted by E_{λ_i} is subspace spanned by all eigenvectors of A corresponding to λ_i
- The **geometric multiplicity** of λ_i is the dimension of E_{λ_i}
- The **spectrum** of A is the set of all eigenvalues of A



Example 2.4.5

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix} \quad (2.15)$$

$$\begin{aligned} \Rightarrow p_A(\lambda) &= \det(A - \lambda I_3) \\ &= \det \left(\begin{bmatrix} 2-\lambda & 1 & 0 \\ 0 & 2-\lambda & 0 \\ 0 & 0 & 3-\lambda \end{bmatrix} \right) \\ &= (2-\lambda)^2(3-\lambda) \end{aligned} \quad (2.16)$$

The **spectrum** of A is $\{2, 3\}$

- With $\lambda = 2$

The **algebraic multiplicity** is 2

$$\begin{aligned} (A - 2I_3)v &= 0 \\ \Rightarrow \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \Rightarrow \begin{cases} y = 0 \\ 0 = 0 \\ z = 0 \end{cases} \\ \Rightarrow v &= \begin{bmatrix} x \\ 0 \\ 0 \end{bmatrix} \text{ for all } x \in \mathbb{R} \end{aligned} \quad (2.17)$$

- With $\lambda = 3$

The **algebraic multiplicity** is 1

$$\begin{aligned} (A - 3I_3)v &= 0 \\ \Rightarrow \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \\ \Rightarrow \begin{cases} -x + y = 0 \\ -y = 0 \\ 0 = 0 \end{cases} \\ \Rightarrow \begin{cases} x = 0 \\ y = 0 \end{cases} \\ \Rightarrow v &= \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} \text{ for all } z \in \mathbb{R} \end{aligned} \quad (2.18)$$

Theorem 2.4.6

For $A \in \mathbb{R}^{n \times n}$

$$\det(A) = \prod_{i=1}^n \lambda_i \quad (2.19)$$

$$\operatorname{tr}(A) = \sum_{i=1}^n \lambda_i \quad (2.20)$$

